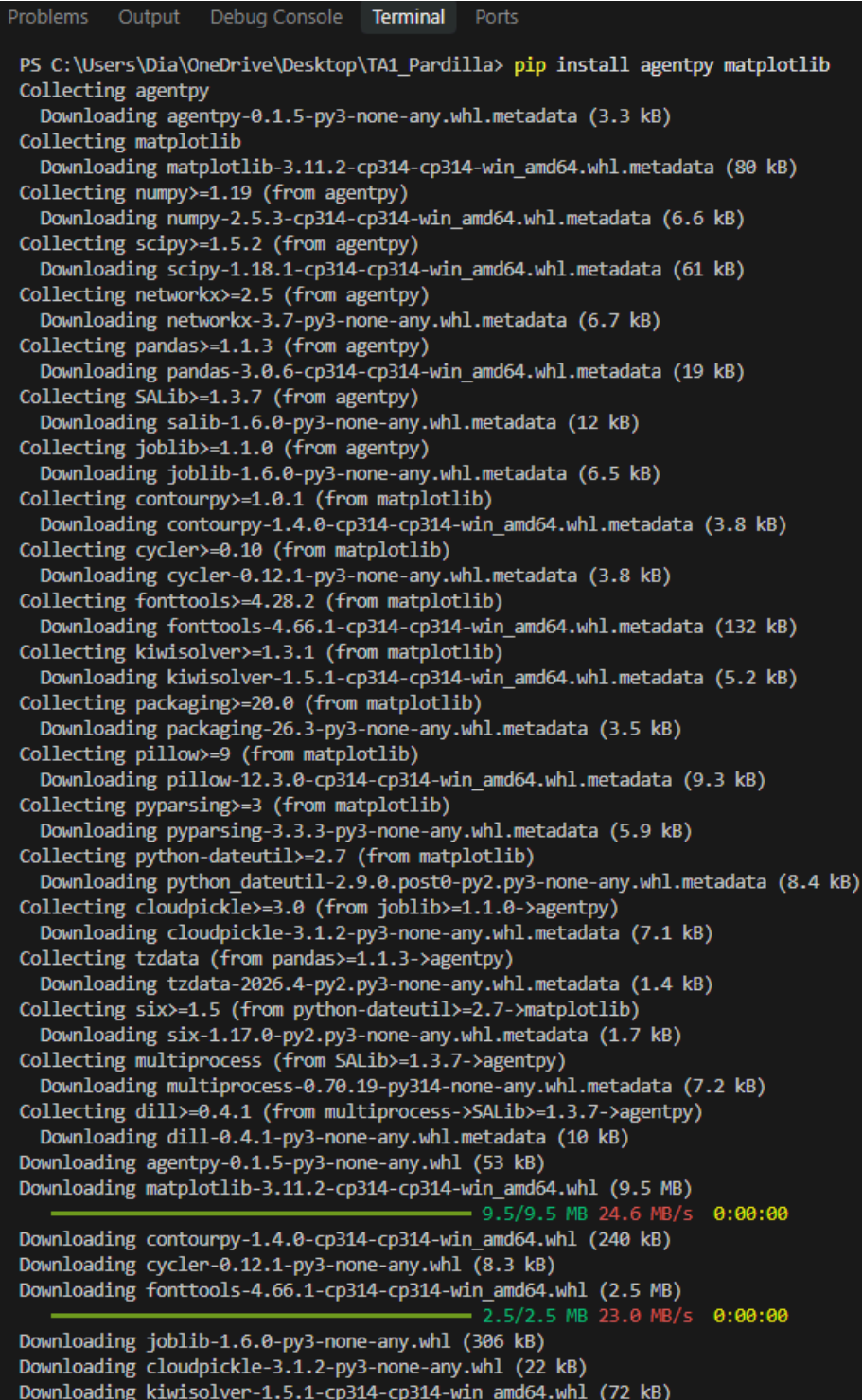


To demonstrate originality, this document provides complete, step-by-step documentation and corresponding screenshots taken throughout the implementation of Technical Assessment 1.

STEP 1:

1. Downloading the agentpy and matplotlib.



```
Problems Output Debug Console Terminal Ports
PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> pip install agentpy matplotlib
Collecting agentpy
  Downloading agentpy-0.1.5-py3-none-any.whl.metadata (3.3 kB)
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl.metadata (80 kB)
Collecting numpy>=1.19 (from agentpy)
  Downloading numpy-2.5.3-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting scipy>=1.5.2 (from agentpy)
  Downloading scipy-1.18.1-cp314-cp314-win_amd64.whl.metadata (61 kB)
Collecting networkx>=2.5 (from agentpy)
  Downloading networkx-3.7-py3-none-any.whl.metadata (6.7 kB)
Collecting pandas>=1.1.3 (from agentpy)
  Downloading pandas-3.0.6-cp314-cp314-win_amd64.whl.metadata (19 kB)
Collecting SALib>=1.3.7 (from agentpy)
  Downloading salib-1.6.0-py3-none-any.whl.metadata (12 kB)
Collecting joblib>=1.1.0 (from agentpy)
  Downloading joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl.metadata (132 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl.metadata (5.2 kB)
Collecting packaging>=20.0 (from matplotlib)
  Downloading packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Collecting python-dateutil>=2.7 (from matplotlib)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting cloudpickle>=3.0 (from joblib>=1.1.0->agentpy)
  Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting tzdata (from pandas>=1.1.3->agentpy)
  Downloading tzdata-2026.4-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.7->matplotlib)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting multiprocessing (from SALib>=1.3.7->agentpy)
  Downloading multiprocessing-0.70.19-py314-none-any.whl.metadata (7.2 kB)
Collecting dill>=0.4.1 (from multiprocessing->SALib>=1.3.7->agentpy)
  Downloading dill-0.4.1-py3-none-any.whl.metadata (10 kB)
Downloading agentpy-0.1.5-py3-none-any.whl (53 kB)
Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl (9.5 MB)
 9.5/9.5 MB 24.6 MB/s 0:00:00
Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl (240 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl (2.5 MB)
 2.5/2.5 MB 23.0 MB/s 0:00:00
Downloading joblib-1.6.0-py3-none-any.whl (306 kB)
Downloading cloudpickle-3.1.2-py3-none-any.whl (22 kB)
Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl (72 kB)
```

2. Copy pasting the code from the main document, translating into a more Visual Studio Code appropriate code, then running the file.

```
TA1_Pardilla.py X
TA1_Pardilla.py > RandomWalkModel > setup
1  import random
2  import agentpy as ap
3  import matplotlib.animation as animation
4  import matplotlib.pyplot as plt
5
6  # ----- 1. Console Inputs -----
7  num_agents = int(input("Enter number of agents: "))
8  grid_size = int(input("Enter grid size (e.g., 10 for 10x10): "))
9  num_steps = int(input("Enter number of steps: "))
10
11
12 # ----- 2. Agent Definition -----
13 class PathTrackingWalker(ap.Agent):
14
15     def setup(self):
16         self.trajectory = []
17
18     def step(self):
19         # Rule change: Weighted drift (40% right, 20% left, up, down)
20         moves = [(1, 0), (-1, 0), (0, 1), (0, -1)]
21         weights = [0.40, 0.20, 0.20, 0.20]
22         direction = random.choices(moves, weights=weights, k=1)[0]
23
24         x, y = self.position
25         new_x = max(0, min(self.model.p.grid_size[0] - 1, x + direction[0]))
26         new_y = max(0, min(self.model.p.grid_size[1] - 1, y + direction[1]))
27
28         self.position = (new_x, new_y)
29         self.trajectory.append(self.position)
30
31
32 # ----- 3. Model Definition -----
33 class RandomWalkModel(ap.Model):
34
35     def setup(self):
36         self.agents = ap.AgentList(self, self.p.agents, PathTrackingWalker)
37
38         for agent in self.agents:
39             init_pos = (
40                 random.randint(0, self.p.grid_size[0] - 1),
41                 random.randint(0, self.p.grid_size[1] - 1),
```

```

41         random.randrange(0, self.p.grid_size[1] - 1))
42     )
43     agent.position = init_pos
44     agent.trajectory.append(init_pos)
45
46     self.grid = ap.Grid(self, self.p.grid_size, torus=False)
47     self.grid.add_agents(self.agents)
48
49     def step(self):
50         self.agents.step()
51
52     def end(self):
53         print("\n" + "=" * 45)
54         print("--- SIMULATION COMPLETE: FINAL POSITIONS ---")
55         print("=" * 45)
56         for i, agent in enumerate(self.agents, 1):
57             print(
58                 f"Agent {i:02d} | Final Position: {agent.position} | Total Points:"
59                 f" {len(agent.trajectory)}"
60             )
61         print("=" * 45 + "\n")
62
63
64     # ----- 4. Model Setup -----
65     parameters = {
66         "agents": num_agents,
67         "grid_size": (grid_size, grid_size),
68         "steps": num_steps,
69     }
70
71     model = RandomWalkModel(parameters)
72     model.setup()
73
74     # ----- 5. Interactive Visualizer -----
75     fig, ax = plt.subplots(figsize=(6, 6))
76
77
78     def update(frame):

```

```

79     if frame > 0:
80         model.step()
81
82     ax.clear()
83     ax.set_xlim(-0.5, model.p.grid_size[0] - 0.5)
84     ax.set_ylim(-0.5, model.p.grid_size[1] - 0.5)
85     ax.set_xticks(range(model.p.grid_size[0]))
86     ax.set_yticks(range(model.p.grid_size[1]))
87     ax.grid(True, linestyle="--", alpha=0.5)
88     ax.set_title(f"Step {frame}/{model.p.steps} - Path Tracing")
89
90     # Draw movement trails
91     for agent in model.agents:
92         xs, ys = zip(*agent.trajectory)
93         ax.plot(xs, ys, alpha=0.5, linewidth=1.5)
94
95     # Draw current agent positions
96     cx = [agent.position[0] for agent in model.agents]
97     cy = [agent.position[1] for agent in model.agents]
98     scat = ax.scatter(cx, cy, s=120, c="crimson", edgecolors="black", zorder=3)
99
100    if frame == model.p.steps:
101        model.end()
102
103    return (scat,)
104
105
106    ani = animation.FuncAnimation(
107        fig, update, frames=model.p.steps + 1, repeat=False, interval=200
108    )
109
110    plt.tight_layout()
111    plt.show()

```

3. Running the main code and answering what is asked:

- a. Number of Agents: 5
- b. Grid Size: 10
- c. Number of Steps: 20

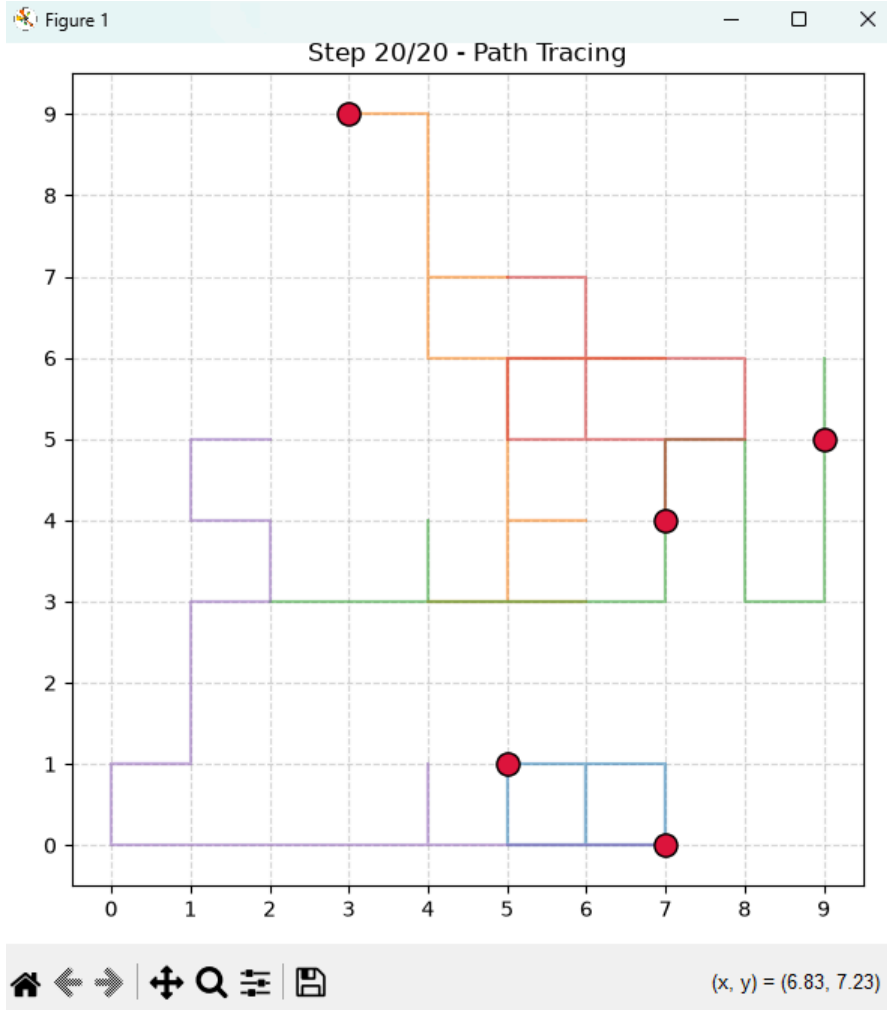
```

PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> & C:\Users\Dia\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/Dia/OneDrive/Desktop/TA1_Pardilla/TA1_Pardilla.py
Enter number of agents: 5
Enter grid size (e.g., 10 for 10x10): 10
Enter number of steps: 20

```

4. Interactive Simulation and the simulation details.

```
=====
--- SIMULATION COMPLETE: FINAL POSITIONS ---
=====
Agent 01 | Final Position: (5, 1) | Total Points: 21
Agent 02 | Final Position: (3, 9) | Total Points: 21
Agent 03 | Final Position: (9, 5) | Total Points: 21
Agent 04 | Final Position: (7, 4) | Total Points: 21
Agent 05 | Final Position: (7, 0) | Total Points: 21
=====
```



STEP 2:

1. Understanding the Agent Behavior

```
# ----- 2. Agent Definition -----  
class PathTrackingWalker(ap.Agent):  
  
    def setup(self):  
        self.trajectory = []  
  
    def step(self):  
        # Rule change: Weighted drift (40% right, 20% left, up, down)  
        moves = [(1, 0), (-1, 0), (0, 1), (0, -1)]  
        weights = [0.40, 0.20, 0.20, 0.20]  
        direction = random.choices(moves, weights=weights, k=1)[0]  
  
        x, y = self.position  
        new_x = max(0, min(self.model.p.grid_size[0] - 1, x + direction[0]))  
        new_y = max(0, min(self.model.p.grid_size[1] - 1, y + direction[1]))  
  
        self.position = (new_x, new_y)  
        self.trajectory.append(self.position)
```

NO STEP 3 BECAUSE IT IS FOR OBSERVING THE SIMULATION.

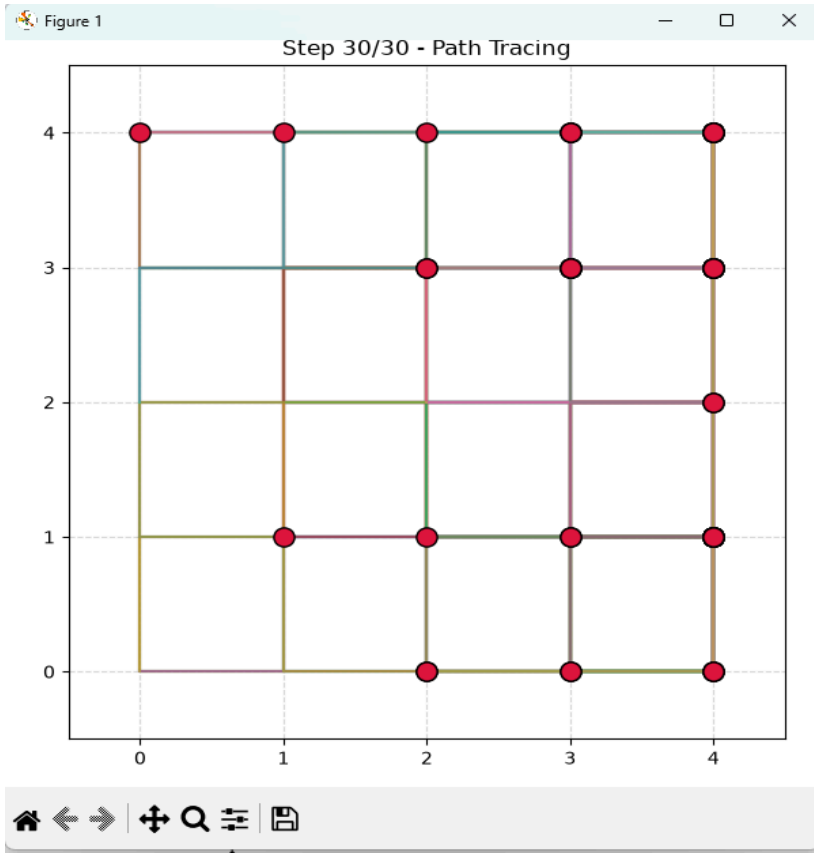
STEP 4:

1. Experimenting with the parameters.
 - a. Experiment 1 (Unmodified Code):
 - i. Number of Agents: 50
 - ii. Grid Size: 5
 - iii. Number of Steps: 30

```
PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> & C:\Users\Dia\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/Dia/OneDrive/Desktop/TA1_Pardilla/TA1_Pardilla.py
Enter number of agents: 50
Enter grid size (e.g., 10 for 10x10): 5
Enter number of steps: 30
```

```
=====
--- SIMULATION COMPLETE: FINAL POSITIONS ---
=====
Agent 01 | Final Position: (3, 1) | Total Points: 31
Agent 02 | Final Position: (4, 4) | Total Points: 31
Agent 03 | Final Position: (2, 0) | Total Points: 31
Agent 04 | Final Position: (4, 4) | Total Points: 31
Agent 05 | Final Position: (1, 1) | Total Points: 31
Agent 06 | Final Position: (4, 1) | Total Points: 31
Agent 07 | Final Position: (4, 0) | Total Points: 31
Agent 08 | Final Position: (4, 1) | Total Points: 31
Agent 09 | Final Position: (4, 4) | Total Points: 31
Agent 10 | Final Position: (3, 3) | Total Points: 31
Agent 11 | Final Position: (4, 3) | Total Points: 31
Agent 12 | Final Position: (4, 1) | Total Points: 31
Agent 13 | Final Position: (4, 1) | Total Points: 31
Agent 14 | Final Position: (2, 3) | Total Points: 31
Agent 15 | Final Position: (4, 3) | Total Points: 31
Agent 16 | Final Position: (4, 3) | Total Points: 31
Agent 17 | Final Position: (3, 3) | Total Points: 31
Agent 18 | Final Position: (3, 3) | Total Points: 31
Agent 19 | Final Position: (1, 4) | Total Points: 31
Agent 20 | Final Position: (2, 4) | Total Points: 31
Agent 21 | Final Position: (4, 1) | Total Points: 31
Agent 22 | Final Position: (4, 3) | Total Points: 31
Agent 23 | Final Position: (0, 4) | Total Points: 31
Agent 24 | Final Position: (4, 0) | Total Points: 31
Agent 25 | Final Position: (2, 0) | Total Points: 31
Agent 26 | Final Position: (4, 3) | Total Points: 31
Agent 27 | Final Position: (4, 0) | Total Points: 31
Agent 28 | Final Position: (3, 4) | Total Points: 31
Agent 29 | Final Position: (2, 3) | Total Points: 31
```

Agent 30	Final Position: (4, 1)	Total Points: 31
Agent 31	Final Position: (4, 3)	Total Points: 31
Agent 32	Final Position: (4, 1)	Total Points: 31
Agent 33	Final Position: (4, 1)	Total Points: 31
Agent 34	Final Position: (3, 3)	Total Points: 31
Agent 35	Final Position: (4, 4)	Total Points: 31
Agent 36	Final Position: (4, 2)	Total Points: 31
Agent 37	Final Position: (4, 2)	Total Points: 31
Agent 38	Final Position: (2, 1)	Total Points: 31
Agent 39	Final Position: (3, 1)	Total Points: 31
Agent 40	Final Position: (3, 0)	Total Points: 31
Agent 41	Final Position: (4, 4)	Total Points: 31
Agent 42	Final Position: (4, 3)	Total Points: 31
Agent 43	Final Position: (3, 4)	Total Points: 31
Agent 44	Final Position: (4, 1)	Total Points: 31
Agent 45	Final Position: (3, 3)	Total Points: 31
Agent 46	Final Position: (3, 0)	Total Points: 31
Agent 47	Final Position: (4, 1)	Total Points: 31
Agent 48	Final Position: (4, 3)	Total Points: 31
Agent 49	Final Position: (3, 4)	Total Points: 31
Agent 50	Final Position: (4, 4)	Total Points: 31
=====		

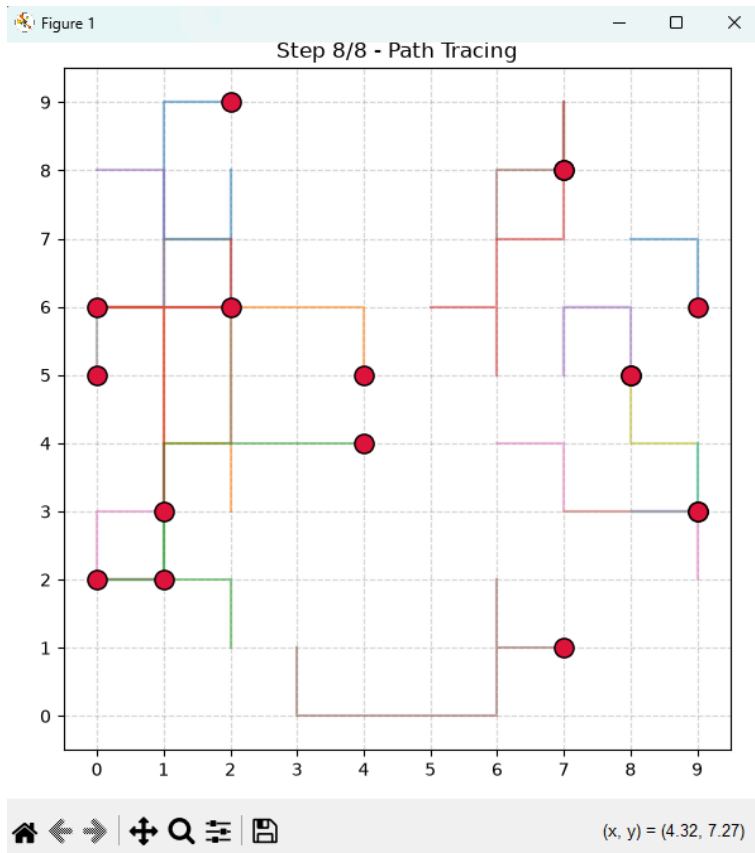


b. Experiment 2 (Unmodified Code):

- i. Number of Agents: 17
- ii. Grid Size: 10
- iii. Number of Steps: 8

```
PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> & C:\Users\Dia\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/Dia/OneDrive/Desktop/TA1_Pardilla/TA1_Pardilla.py
Enter number of agents: 17
Enter grid size (e.g., 10 for 10x10): 10
Enter number of steps: 8
```

```
=====
--- SIMULATION COMPLETE: FINAL POSITIONS ---
=====
Agent 01 | Final Position: (2, 9) | Total Points: 9
Agent 02 | Final Position: (4, 5) | Total Points: 9
Agent 03 | Final Position: (0, 2) | Total Points: 9
Agent 04 | Final Position: (7, 8) | Total Points: 9
Agent 05 | Final Position: (0, 6) | Total Points: 9
Agent 06 | Final Position: (7, 8) | Total Points: 9
Agent 07 | Final Position: (1, 2) | Total Points: 9
Agent 08 | Final Position: (0, 5) | Total Points: 9
Agent 09 | Final Position: (8, 5) | Total Points: 9
Agent 10 | Final Position: (9, 3) | Total Points: 9
Agent 11 | Final Position: (9, 6) | Total Points: 9
Agent 12 | Final Position: (2, 6) | Total Points: 9
Agent 13 | Final Position: (4, 4) | Total Points: 9
Agent 14 | Final Position: (1, 3) | Total Points: 9
Agent 15 | Final Position: (8, 5) | Total Points: 9
Agent 16 | Final Position: (7, 1) | Total Points: 9
Agent 17 | Final Position: (9, 3) | Total Points: 9
=====
```



- c. Experiment 3 (Modified Code):
- i. Number of Agents: 6
 - ii. Grid Size: 2
 - iii. Number of Steps: 15

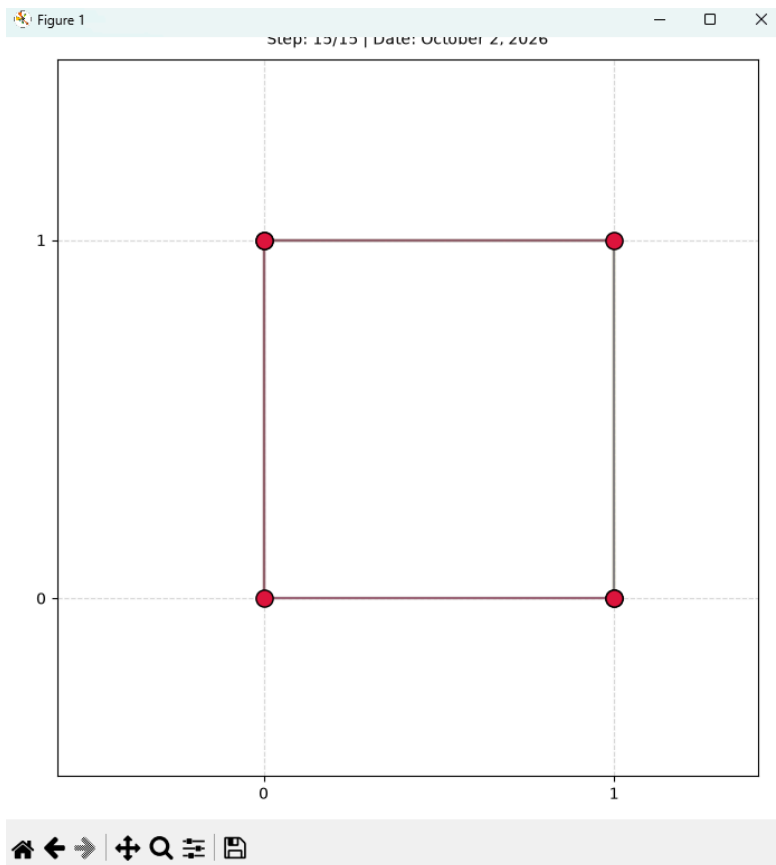
```
PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> C:\Users\Dia\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Dia/OneDrive/Desktop/TA1_Pardilla/TA1 Modified_Pardilla.py"
```

```
GridDrift: Multi-Agent Random Walk Simulation  
CS0065 (Intelligent Systems) | Section: AN41  
Programmed by: Dianna Francesca M. Pardilla  
Date: October 2, 2026
```

```
Enter number of agents: 6  
Enter grid size (e.g., 10 for 10x10): 2  
Enter number of steps: 15
```

```
=====
                        FINAL POSITIONS SUMMARY
Programmed by: Dianna Francesca M. Pardilla | AN41
=====

Agent 01 | Start: (0, 0) | Final: (0, 0) | Total Steps: 15
Agent 02 | Start: (1, 0) | Final: (1, 0) | Total Steps: 15
Agent 03 | Start: (0, 1) | Final: (0, 1) | Total Steps: 15
Agent 04 | Start: (1, 1) | Final: (1, 0) | Total Steps: 15
Agent 05 | Start: (1, 0) | Final: (1, 1) | Total Steps: 15
Agent 06 | Start: (0, 1) | Final: (0, 1) | Total Steps: 15
=====
```

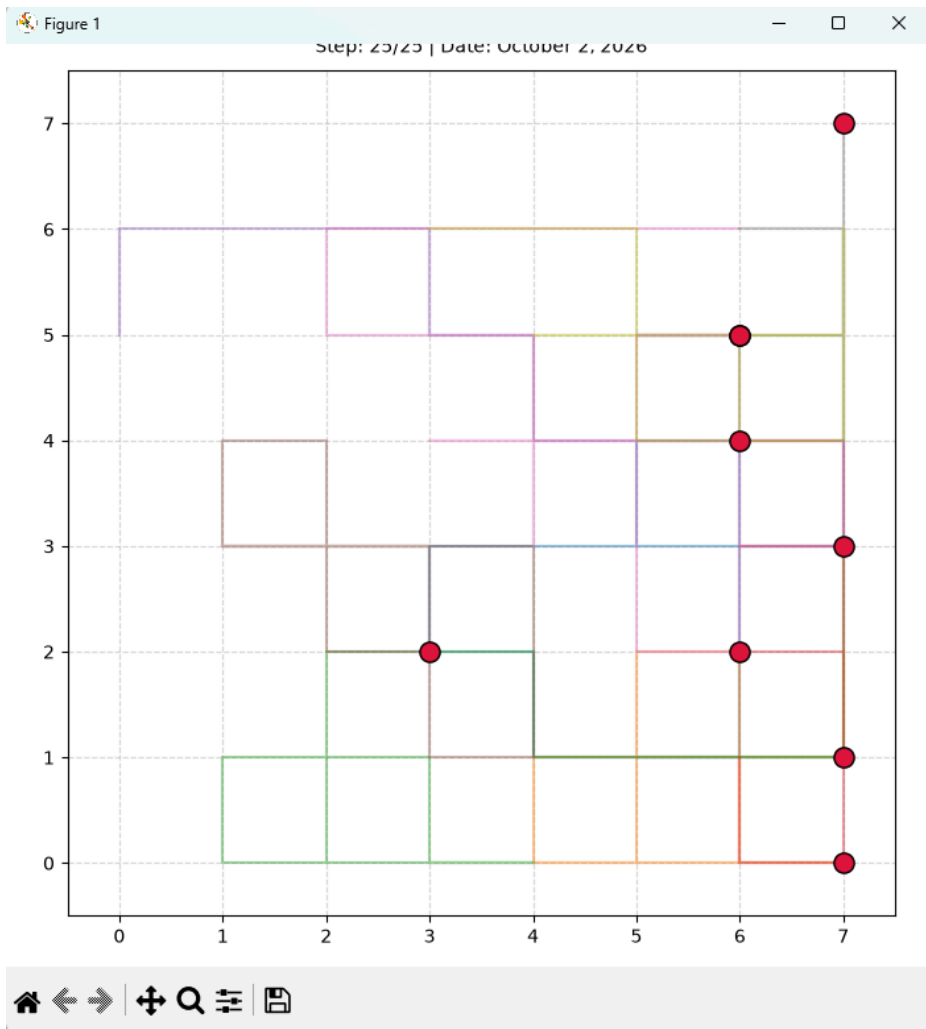


- d. Experiment 4 (Modified Code):
- i. Number of Agents: 9
 - ii. Grid Size: 8
 - iii. Number of Steps: 25

```
PS C:\Users\Dia\OneDrive\Desktop\TA1_Pardilla> & C:\Users\Dia\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/Dia/OneDrive/Desktop/TA1_Pardilla/TA1 Modified_Pardilla.py"
GridDrift: Multi-Agent Random Walk Simulation
CS0065 (Intelligent Systems) | Section: AN41
Programmed by: Dianna Francesca M. Pardilla
Date: October 2, 2026
~~~~~

Enter number of agents: 9
Enter grid size (e.g., 10 for 10x10): 8
Enter number of steps: 25
```

FINAL POSITIONS SUMMARY			
Programmed by: Dianna Francesca M. Pardilla AN41			
Agent 01	Start: (4, 1)	Final: (7, 1)	Total Steps: 25
Agent 02	Start: (5, 2)	Final: (7, 0)	Total Steps: 25
Agent 03	Start: (2, 0)	Final: (7, 3)	Total Steps: 25
Agent 04	Start: (7, 0)	Final: (6, 2)	Total Steps: 25
Agent 05	Start: (0, 6)	Final: (6, 5)	Total Steps: 25
Agent 06	Start: (1, 3)	Final: (3, 2)	Total Steps: 25
Agent 07	Start: (6, 6)	Final: (6, 5)	Total Steps: 25
Agent 08	Start: (7, 6)	Final: (7, 7)	Total Steps: 25
Agent 09	Start: (3, 6)	Final: (6, 4)	Total Steps: 25



2. Modified Code

```
TA1 Modified_Pardilla.py > ...
1  import random
2  import agentpy as ap
3  import matplotlib.animation as animation
4  import matplotlib.pyplot as plt
5
6  APP_TITLE = "GridDrift: Multi-Agent Random Walk Simulation"
7
8  # < CONSOLE BANNER & HEADER >
9  print(f" {APP_TITLE} ")
10 print(" CS0065 (Intelligent Systems) | Section: AN41")
11 print(" Programmed by: Dianna Francesca M. Pardilla")
12 print(" Date: October 2, 2026")
13 print("~" * 60 + "\n")
14
15 # 1. < USER INPUTS >
16 num_agents = int(input("Enter number of agents: "))
17 grid_size = int(input("Enter grid size (e.g., 10 for 10x10): "))
18 num_steps = int(input("Enter number of steps: "))
19
20 # 2. < AGENT DEFINITION (LAB TASKS) >
21 class CustomWalker(ap.Agent):
22
23     def setup(self):
24         self.trajectory = []
25
26     def step(self):
27         # Rule change: Weighted drift (40% right, 20% left, up, down)
28         moves = [(1, 0), (-1, 0), (0, 1), (0, -1)]
29         weights = [0.40, 0.20, 0.20, 0.20]
30         direction = random.choices(moves, weights=weights, k=1)[0]
31
32         x, y = self.position
33
34         # Keep agents inside grid limits
35         bounded_x = max(
36             0, min(self.model.p.grid_size[0] - 1, x + direction[0])
37         ) #cite: 1]
38         bounded_y = max(
39             0, min(self.model.p.grid_size[1] - 1, y + direction[1])
40         ) #cite: 1]
41
42         self.position = (bounded_x, bounded_y)
43         self.trajectory.append(self.position)
44
```

```

TA1 Modified_Pardilla.py > ...
45 # 3. < MODEL DEFINITION >
46 class RandomWalkModel(ap.Model):
47
48     def setup(self):
49         self.agents = ap.AgentList(self, self.p.agents, CustomWalker)
50
51         for agent in self.agents:
52             init_pos = (
53                 random.randint(0, self.p.grid_size[0] - 1),
54                 random.randint(0, self.p.grid_size[1] - 1),
55             )
56             agent.position = init_pos
57             agent.trajectory.append(init_pos)
58
59         self.grid = ap.Grid(self, self.p.grid_size, torus=False)
60         self.grid.add_agents(self.agents)
61
62     def step(self):
63         self.agents.step()
64
65     def end(self):
66         print("\n" + "~" * 60)
67         print("                FINAL POSITIONS SUMMARY                ")
68         print(" Programmed by: Dianna Francesca M. Pardilla | AN41 ")
69         print("~" * 60 + "\n")
70         for i, agent in enumerate(self.agents, 1):
71             start_coord = agent.trajectory[0]
72             final_coord = agent.position
73             print(
74                 f"Agent {i:02d} | Start: {start_coord} | Final: {final_coord} | Total"
75                 f" Steps: {len(agent.trajectory) - 1}"
76             )
77         print("=" * 60 + "\n")
78
79 # 4. < EXECUTION & ANIMATION >
80 parameters = {
81     "agents": num_agents,
82     "grid_size": (grid_size, grid_size),
83     "steps": num_steps,
84 }
85
86 model = RandomWalkModel(parameters)
87 model.setup()
88

```

TA1 Modified_Pardilla.py > ...

```
89 fig, ax = plt.subplots(figsize=(7, 7))
90
91
92 def update(frame):
93     if frame > 0:
94         model.step()
95
96     ax.clear()
97     ax.set_xlim(-0.5, model.p.grid_size[0] - 0.5)
98     ax.set_ylim(-0.5, model.p.grid_size[1] - 0.5)
99     ax.set_xticks(range(model.p.grid_size[0]))
100    ax.set_yticks(range(model.p.grid_size[1]))
101    ax.grid(True, linestyle="--", alpha=0.5)
102
103    # Clean title header on the plot window
104    ax.set_title(
105        f"{APP_TITLE}\n"
106        "Programmed by: Dianna Francesca M. Pardilla | CS0065 - AN41\n"
107        f"Step: {frame}/{model.p.steps} | Date: October 2, 2026",
108        fontsize=10,
109        pad=10,
110    )
111
112    # Draw movement trails
113    for agent in model.agents:
114        xs, ys = zip(*agent.trajectory)
115        ax.plot(xs, ys, alpha=0.45, linewidth=1.5)
116
117    # Draw current positions
118    cx = [agent.position[0] for agent in model.agents]
119    cy = [agent.position[1] for agent in model.agents]
120    scat = ax.scatter(
121        cx, cy, s=120, c="crimson", edgecolors="black", zorder=3, label="Agents"
122    )
123
124    if frame == model.p.steps:
125        model.end()
126
127    return (scat,)
128
129
130 ani = animation.FuncAnimation(
131     fig, update, frames=model.p.steps + 1, repeat=False, interval=200
```

```
132 )
133
134 plt.tight_layout()
135 plt.show()
```